

KIRCHHOFF’S LAW OF THERMAL EMISSION WITH BUILT-IN UV/IR CUTOFFS FROM HQIV’S DISCRETE NULL LATTICE

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ABSTRACT. Result. Kirchhoff’s law of thermal emission is recovered from two HQIV axioms — a discrete null lattice with Planck cell size and informational monogamy ($\alpha = 3/5$) — with a *finite* mode count at every shell, given in closed form by the stars-and-bars identity

$$\mathcal{N}(m) = (m+2)(m+1), \quad \sum_{m=0}^n \mathcal{N}(m) = \frac{1}{3}(n+1)(n+2)(n+3).$$

The resulting Planck spectrum is a *finite sum* between an ultraviolet cutoff at $m = 0$ (the Planck pole, $\omega_{\text{Pl}} = T_{\text{Pl}}$) and an arbitrary infrared cutoff at $m = M$ (the lock-in reference shell). No ultraviolet catastrophe arises, no renormalisation is performed, and no external blackbody calibration is imposed: the mode budget is *finite by construction*, not by regularisation.

Sourcing of the dimensionful inputs. The proton mass at the lock-in shell $m = 4$ uses the theory’s internal constituent-plus-binding readout, with the observed value $M_p = 938.272$ MeV serving as the *active scale witness* (`proton_lockin`; see `AGENTS/ASSUMPTIONS.md §7g` for the witness policy). The wall-clock age $t_{\text{wall}} = 51.2$ Gyr is an *output* of the HQIV-modified CLASS implementation — the `class_hqiv` tree combining a discrete null-lattice background with the ADM lapse $N(t) = 1 + \Phi + \varphi t$ and φ -dependent evolution (`class_hqiv_patches/source/hqiv.c::hqiv_wall_clock_compression`) — not a free parameter. The only externally measured dimensionful input used downstream of these is FIRAS $T_{\text{CMB}} = 2.7255$ K.

One quantitative prediction under the chosen witness. With `proton_lockin` fixing the mass/unit chart, all other dimensionful quantities — CODATA $1/\alpha$, CMB horizons, cosmic birefringence — become *predictions or comparison layers*. We exhibit one such prediction here:

- **Isotropic cosmic birefringence at present epoch.** The birefringence calculation uses the HQIV wall-clock age above together with the lattice mode count $\mathcal{N}(m)$ under the propagation-shell identification (§7), giving $\beta_{\text{HQIV}} = 0.379^\circ$. This agrees with Planck PR4 EB ($\beta_{\text{obs}} = 0.342^\circ \pm 0.094^\circ$, Eskilt & Komatsu 2022) within -0.40σ . Substituting HQIV’s apparent age 13.8 Gyr for the wall-clock age gives 0.103° (-2.55σ): the prediction *discriminates* HQIV’s two internal time outputs.

Status. The derivation chain is mechanised sorry-free against mathlib in `Hqiv/Geometry/OctonionicLightCone.lean`, `Hqiv/Physics/HorizonBlackbodySpectrum.lean`, and `Hqiv/Cosmology/CosmologicalShellLadder.lean`. The Friedmann radiation-era scaling $H \propto T^2$ is also recovered, not imported: it follows from the quadratic asymptotic form of $\mathcal{N}(m)$ under the propagation-shell identification.

1. SCOPE AND CLAIM DISCIPLINE

This paper presents a derivation of Kirchhoff’s law of thermal emission inside HQIV. We do *not* reprove the classical quantum statistical mechanics; we replace its mode-density input with HQIV’s finite stars-and-bars count and show that the resulting blackbody machinery (i) is finite at every step, (ii) admits explicit UV and IR cutoffs by construction, and (iii) supports one downstream quantitative prediction (β at present epoch) that survives the data, under HQIV’s own scale-witness discipline.

- (1) **Two axioms.** Discrete null lattice with Planck cell size; informational monogamy fixing $\alpha = 3/5$ via the hockey-stick identity. These two are the only HQIV combinatorial inputs.
- (2) **Finite by construction.** All spectral sums are $\sum_{m \in [m_{\text{UV}}, m_{\text{IR}}]}$ over finite shell intervals in $\mathbb{N}$. No limits, no analytic continuation, no cutoff-removal procedure.

- (3) **Single active scale witness.** HQIV’s standing discipline (`AGENTS/ASSUMPTIONS.md §7g`) admits exactly one dimensionful witness per solve. Throughout this paper we use the default, `proton_lockin`: the proton mass at the lock-in shell $m = 4$ uses the theory’s internal constituent-plus-binding readout, with the observed value $M_p = 938.272$ MeV serving as the active scale witness that pins the mass/unit chart. The shell index $m = 4$ is the QCD-onset identification (not a free parameter), and the constituent-plus-binding closure at that shell is a non-trivial HQIV calculation, but the numerical scale of 938.272 MeV is the *anchor*, not an independent two-axiom output. See §6 for the honest accounting.
- (4) **Wall-clock age from `class_hqiv`.** The age $t_{\text{wall}} = 51.2$ Gyr used in the β prediction is an output of the HQIV-modified CLASS implementation (the `class_hqiv` tree, combining a discrete null-lattice background with the ADM lapse $N(t) = 1 + \Phi + \varphi t$ and φ -dependent evolution; function `hqiv_wall_clock_compression` in `class_hqiv_patches/source/hqiv.c`), not a free parameter. Its Lean snapshot is `age_wall_clock_Gyr_paper`.
- (5) **Predictions vs anchor.** Under `proton_lockin`, observables such as CODATA $1/\alpha$, cosmic birefringence β , and CMB horizon scales are *predictions or comparison layers*, not simultaneous anchors. This paper exhibits β as one such prediction.
- (6) **β is non-adjustable under the chosen witness.** Given the anchor, β_{HQIV} uses only the HQIV stars-and-bars count, the HQIV wall-clock age from `class_hqiv`, and FIRAS T_{CMB} . No birefringence-side anchor and no Friedmann equation enter the calculation. The internal discriminator wall-clock vs apparent age is a single-shot test.

2. BACKGROUND: CLASSICAL KIRCHHOFF AND THE NORMALISATION PROBLEM

Kirchhoff’s law of thermal emission states that, at thermodynamic equilibrium, the spectral radiance emitted per mode equals the spectral radiance absorbed per mode — the universal blackbody spectrum. Two ingredients enter:

- (1) Detailed balance per mode (microscopic reversibility);
- (2) A mode-density per frequency interval $dN/d\nu$.

In the standard derivation, (i) is statistical-mechanical and robust; (ii) is the problematic input. Rayleigh–Jeans gives $dN/d\nu \propto \nu^2$ in a 3D cavity, unbounded above, and so the classical mean-energy multiplication $k_B T \cdot dN/d\nu$ diverges as $\nu \rightarrow \infty$ — the ultraviolet catastrophe. Planck’s quantum cure provides $\langle E \rangle = h\nu/(e^{h\nu/k_B T} - 1)$, which exponentially suppresses high- ν contributions and yields a finite total energy.

This works numerically. But the cure is asymmetric: *the high-frequency cutoff is contributed by occupation suppression, not by mode counting*. In particular, nothing about the classical derivation prevents the existence of modes at arbitrarily high frequency; one simply trusts the Bose factor to render them harmless. This is the source of the normalisation question — if the mode count itself is unbounded, any quantum-gravity or trans-Planckian completion has to act on the mode list, not just on the occupation.

HQIV makes the mode count finite at the start.

3. HQIV MODE-COUNT AXIOM

Definition 3.1 (Discrete null-lattice mode count). *Let $\mathcal{N}(m)$ be the count of new Planck-scale lattice cells at temperature-ladder depth $m \in \mathbb{N}$. HQIV’s discrete null-lattice axiom gives*

$$(1) \quad \mathcal{N}(m) = (m+2)(m+1).$$

This is the integer-solution count for $x + y + z = m$, $x, y, z \geq 0$, twice the binomial $\binom{m+2}{2}$.

Proposition 3.2 (Hockey-stick cumulative count).

$$(2) \quad \sum_{m'=0}^n \mathcal{N}(m') = \frac{1}{3}(n+1)(n+2)(n+3).$$

Proof. By induction on n , using (1). The Lean witness is `cumLatticeSimplexCount_hockey_stick` in `Hqiv/Geometry/OctonionicLightCone.lean`, which states $3 \sum_{m'=0}^n \mathcal{N}(m') = (n+1)(n+2)(n+3)$. $\square$

Corollary 3.3 (Both cutoffs finite). *For any $0 \leq m_{\text{UV}} \leq m_{\text{IR}} < \infty$,*

$$\sum_{m \in [m_{\text{UV}}, m_{\text{IR}}]} \mathcal{N}(m) < \infty.$$

This is the structural fact that lets the entire spectral machinery be a finite sum, regardless of the temperature and without any regularisation procedure.

4. HQIV TEMPERATURE AND FREQUENCY LADDERS

The HQIV harmonic ladder fixes the shell-indexed frequency at:

$$(3) \quad \omega_m = \frac{T_{\text{Pl}}}{m+1}, \quad m \in \mathbb{N},$$

with $\omega_0 = T_{\text{Pl}}$ at the Planck pole (highest frequency, no UV catastrophe) and $\omega_m \rightarrow 0$ as $m \rightarrow \infty$ (IR limit, only $\mathcal{N}(m) \sim m^2$ modes each). Equivalently, the temperature ladder is $T(m) = T_{\text{Pl}}/(m+1)$ — this is `Hqiv.Geometry.AuxiliaryField.T` in the repository.

5. KIRCHHOFF'S LAW ON A FINITE SHELL LIST

We now state Kirchhoff's law in HQIV form. Two ingredients, both finite by construction.

Definition 5.1 (Per-shell Bose occupation). *At temperature $T > 0$ and shell $m \in \mathbb{N}$,*

$$(4) \quad n_B(\omega_m, T) = \frac{1}{\exp(\omega_m/T) - 1}.$$

This is the standard Bose factor on one HQIV mode, formally `Hqiv.Physics.HorizonBlackbodySpectrum.nBose`. Positive and finite at every $m \in \mathbb{N}$ and $T > 0$ (with the divergence at $\omega_m = T \rightarrow 0$ outside the shell-discrete sector).

Definition 5.2 (Per-shell spectral energy).

$$(5) \quad E_m(T) = \mathcal{N}(m) \cdot \omega_m \cdot n_B(\omega_m, T).$$

Formally `Hqiv.Physics.HorizonBlackbodySpectrum.shellSpectralEnergy`. Every factor is positive at every m, T ; no regulariser is applied.

Definition 5.3 (Truncated blackbody energy density). *For any $m_{\text{UV}} \leq m_{\text{IR}}$ in $\mathbb{N}$,*

$$(6) \quad u(T; m_{\text{UV}}, m_{\text{IR}}) = \sum_{m=m_{\text{UV}}}^{m_{\text{IR}}} E_m(T).$$

Formally `Hqiv.Physics.HorizonBlackbodySpectrum.blackbodyEnergyDensity`.

Theorem 5.4 (Finite-mode Kirchhoff law). *For any $T > 0$ and any $m_{\text{UV}} \leq m_{\text{IR}}$ in $\mathbb{N}$, (6) is a finite, positive, monotone sum:*

- (1) $0 \leq u(T; m_{\text{UV}}, m_{\text{IR}}) < \infty$;
- (2) $0 < u(T; m_{\text{UV}}, m_{\text{IR}})$ whenever $m_{\text{UV}} \leq m_{\text{IR}}$;
- (3) monotone in both cutoffs: lowering m_{UV} or raising m_{IR} adds non-negative contributions.

The result is true without any limiting procedure or renormalisation: it holds as a closed-form identity on $\mathbb{N}^2 \times \mathbb{R}_{>0}$.

Proof. (i) is `blackbodyEnergyDensity_nonneg` and finiteness from finite-range $\mathbb{N}$ -sum. (ii) is `blackbodyEnergyDensity_pos_of_le`. (iii) is `blackbodyEnergyDensity_mono_UV` and `blackbodyEnergyDensity_m`. $\square$

Remark 5.5 (The UV catastrophe never appears). *Unlike the classical derivation, no high-frequency divergence exists to suppress. $\mathcal{N}(m)$ is exactly $(m+2)(m+1)$ at the Planck pole $m=0$, namely 2 modes (lattice points $(0,0,1), (0,1,0), (1,0,0)$ scaled to two orientations), and there is no shell above it: $\omega > T_{\text{Pl}}$ does not exist in the HQIV mode list. The Bose factor in (4) additionally provides exponential suppression at fixed $T \ll T_{\text{Pl}}$, so the Planck spectrum shape is preserved everywhere classical spectroscopy probes; but the structural finiteness is independent of any thermal cutoff.*

Remark 5.6 (The IR is also finite). *At fixed temperature T , the Bose factor diverges as $\omega_m \rightarrow 0$ ($m \rightarrow \infty$): $n_B(\omega_m, T) \sim T/\omega_m = T(m+1)/T_{\text{Pl}}$, while $\omega_m \rightarrow 0$. The product $\omega_m n_B(\omega_m, T) \rightarrow T$, finite per mode; multiplying by $\mathcal{N}(m) \sim m^2$ gives a per-shell contribution growing as $m^2 T$. If one took $m_{\text{IR}} \rightarrow \infty$ the sum would diverge, exactly the classical Rayleigh–Jeans IR behavior — but HQIV’s discrete null lattice places a structural cutoff at the lock-in reference shell M , formally `Hqiv.Physics.HorizonBlackbodySpectrum.referenceIRCutoff`, making the IR also finite at the lattice level, not just by stipulation.*

Definition 5.7 (Vacuum zero-point cell).

$$(7) \quad u_0(m_{\text{UV}}, m_{\text{IR}}) = \sum_{m=m_{\text{UV}}}^{m_{\text{IR}}} \frac{\mathcal{N}(m) \omega_m}{2}.$$

Formally `Hqiv.Physics.HorizonBlackbodySpectrum.vacuumZeroPointEnergy`. Also non-negative on every shell interval (`vacuumZeroPointEnergy_nonneg`).

Theorem 5.8 (Decomposition).

$$(8) \quad u_{\text{tot}}(T; m_{\text{UV}}, m_{\text{IR}}) = u_0(m_{\text{UV}}, m_{\text{IR}}) + u(T; m_{\text{UV}}, m_{\text{IR}}).$$

Formally `Hqiv.Physics.HorizonBlackbodySpectrum.blackbodyEnergyDensityWithVacuum`, plus the additive decomposition lemma in the same file. No renormalisation is required to handle the zero-point cell — it is a finite, structural part of the spectrum on a finite shell list.

Summary. Equation (8) is the HQIV statement of Kirchhoff’s law. The standard Stefan–Boltzmann scaling $u(T; 0, M) \sim T^4$ and the Wien displacement bound emerge from (6) via the existing `HorizonBlackbodyStefan` and `HorizonBlackbodyWienDisplacement` modules, and rely only on Theorem 5.4. No regularisation appears in any step.

6. THE SCALE ANCHOR: PROTON MASS AT THE LOCK-IN SHELL $m=4$

We use this section to be explicit about what the proton mass *is* and *is not* doing in this paper, since careless phrasing here can over-claim independence.

Status under `proton_lockin`. Per `AGENTS/ASSUMPTIONS.md §7g`, the HQIV pipeline carries a single active dimensionful scale witness per solve. The default is `proton_lockin`: the proton mass at the lock-in shell $m=4$ uses the theory’s internal constituent-plus-binding readout (described below), with the *observed* value 938.272 MeV serving as the active scale witness that pins the mass/unit chart. Other quantities — CODATA $1/\alpha$, CMB horizons, PDG centrals — are then predictions or comparison layers against this anchor. The Lean enum is `Hqiv.Physics.ScaleWitness`, and the documented rule is: “never wire CODATA α and proton mass and a third cosmology anchor into the same solve without documenting which one is active and which are predictions.”

Shell index $m = 4$: chosen, not derived. The reference shell $m = 4$ is the QCD-onset identification: it is the HQIV temperature-ladder index where colour confinement and the lock-in chart begin. The choice is not a free fit parameter — the hadronic mass content of the universe requires *some* QCD-onset shell, and $m = 4$ is the smallest integer consistent with the small number of post-QCD steps needed to reach the nucleon harmonic. But it is also not a two-axiom output: the two axioms by themselves do not single out $m = 4$.

What *is* non-trivial at $m = 4$. Given the shell choice, the readout

$$(9) \quad M_p^{\text{HQIV}} = M_p^{\text{const}} - E_{\text{bind}}^{\text{shared}} = 938.272 \text{ MeV}$$

(formally `Hqiv.Physics.DerivedNucleonMass.derivedProtonMass`, with `proton_mass_from_shared_harmonics` establishing (9)) packages the nucleon as constituent-energy minus shared binding. The constituent harmonics M_p^{const} are dictated by the mode count (1) at $m = 4$; the shared-binding energy $E_{\text{bind}}^{\text{shared}}$ has its own composite-trace construction. These combinatorial structures are genuine HQIV content, and the closure to a single number 938.272 MeV is a non-trivial constraint on the lock-in shell-set. The Lean constant `Hqiv.Physics.QuarkMetaResonance.protonAnchorMass_MeV` records this as the active scale witness.

Why this is not a “test” in the same sense as β . At the level of the broader theory, the numerical value of M_p^{HQIV} is the chart-pinning anchor: it defines the unit of mass/energy used to interpret every other dimensionful quantity. Calling it a “test” would conflate two distinct roles: (i) the narrow constituent-plus-binding readout at the lock-in shell, which is a genuine HQIV calculation, vs (ii) the position of the lock-in shell on the integer ladder, which is a modelling choice in the broader theory. The honest framing is: M_p is the *anchor*, established once; β is the *prediction*, evaluated against the anchor and against the data.

Cross-check status. Within HQIV’s own discipline, candidate scale-witness alternatives are recorded (`codata_alpha`, `cmb_now`) and exist in the Lean enum. Switching to a different witness is a deliberate operation and would re-pin every dimensionful HQIV output. For this paper all numerics are reported under `proton_lockin`.

7. QUANTITATIVE TEST: COSMIC BIREFRINGENCE $\beta = 0.379^\circ$

This is the section where HQIV makes a single-shot quantitative prediction under the chosen scale witness. The proton-mass anchor of §6 has already pinned the unit chart. The birefringence calculation then combines the HQIV wall-clock age $t_{\text{wall}} = 51.2 \text{ Gyr}$ — an output of the HQIV-modified CLASS implementation (`class_hqiv` tree, ADM lapse $N(t) = 1 + \Phi + \varphi t$, φ -dependent evolution, function `hqiv_wall_clock_compression` in `class_hqiv_patches/source/hqiv.c`) — with the same stars-and-bars mode count $\mathcal{N}(m) = (m+2)(m+1)$ of (1) under the propagation-shell identification. Only one externally measured dimensionful input enters at this stage: FIRAS $T_{\text{CMB}} = 2.7255 \text{ K}$. No birefringence-side anchor enters — in particular, no PR4 calibration — and no Friedmann equation is imposed.

The Planck DR4 EB analysis of Eskilt & Komatsu reports an isotropic cosmic birefringence

$$(10) \quad \beta_{\text{obs}} = 0.342^\circ \pm 0.094^\circ \quad (\text{Planck PR4 NPIPE, EB cross-spectrum}),$$

a $\sim 3.6\sigma$ detection, frequency-independent across the 30–353 GHz Planck sub-bands, isotropic within errors.

Two shell coordinates. HQIV carries two genuinely distinct shell coordinates (§3, §4, and `Hqiv/Cosmology/Cosmology`)

$$(11) \quad m_T(T_{\text{obs}}) = T_{\text{Pl}}/T_{\text{obs}} - 1 \quad (\text{continuous, one index per Planck-frequency step}),$$

$$(12) \quad m_{\text{prop}} = \text{count of HQIV propagation cells crossed since the Planck pole.}$$

A photon’s proper time is zero, so it does not count continuous temperature-ladder positions; the cumulative birefringence formula uses the propagation coordinate.

Propagation-shell coarseness. Identifying one HQIV propagation shell at observation depth m_T with the bundle of new lattice simplices appearing at that depth,

$$(13) \quad \text{Planck cells per propagation shell} = \mathcal{N}(m_T),$$

the propagation-shell offset at the observation event is

$$(14) \quad m_{\text{prop}} = \frac{t_{\text{wall}}/t_{\text{Pl}}}{\mathcal{N}(m_T)} = \frac{t_{\text{wall}}/t_{\text{Pl}}}{(m_T + 2)(m_T + 1)} \approx \frac{t_{\text{wall}}}{t_{\text{Pl}}} \left(\frac{T_{\text{obs}}}{T_{\text{Pl}}} \right)^2.$$

This is the central use of the same stars-and-bars count (1) that delivered the finite-mode Kirchhoff law in §5; we are not introducing a new HQIV input.

Birefringence formula. HQIV's cumulative birefringence observable, proven sorry-free as `Hqiv.Physics.HorizonBL`, is

$$(15) \quad \beta(m_{\text{emit}}, m_{\text{obs}}) = \alpha \log \left(\frac{m_{\text{obs}} + 1}{m_{\text{emit}} + 1} \right),$$

with $\alpha = 3/5$. Specialising to the near-pole reading ($m_{\text{emit}} = 0$ at the Planck pole, $m_{\text{obs}} = m_{\text{prop}}$):

$$(16) \quad \beta_{\text{HQIV}} = \alpha \log(1 + m_{\text{prop}}).$$

Numerics. Inputs: $T_{\text{Pl}} = 1.41679 \times 10^{32}$ K, $T_{\text{CMB}} = 2.7255$ K (FIRAS), $t_{\text{Pl}} = 5.391 \times 10^{-44}$ s, $t_{\text{wall}} = 51.2$ Gyr (output of the `class_hqiv`-modified CLASS background; Lean snapshot `age_wall_clock_Gyr_paper`), $\alpha = 3/5$ (informational monogamy). Then

$$\begin{aligned} m_T &= 5.198 \times 10^{31}, \\ \mathcal{N}(m_T) &= 2.702 \times 10^{63}, \\ t_{\text{wall}}/t_{\text{Pl}} &= 2.997 \times 10^{61}, \\ m_{\text{prop}} &= 1.109 \times 10^{-2}. \end{aligned}$$

$$(17) \quad \beta_{\text{HQIV}} = \frac{3}{5} \log(1.01109) = 6.619 \times 10^{-3} \text{ rad} = \boxed{0.3792^\circ},$$

deviation -0.40σ from (10).

Falsifier: apparent age fails. Substituting the HQIV apparent age 13.8 Gyr (`age_apparent_Gyr_paper`) for t_{wall} : $m_{\text{prop}}^{\text{app}} \approx 3.0 \times 10^{-3}$, $\beta^{\text{app}} \approx 0.103^\circ$, a -2.55σ deviation. The formula *selects* HQIV's wall-clock age over the apparent age — this is a non-adjustable test of the ADM-lapse subsystem against an independent observation.

Observer-position character (PR4 frequency independence). By (16) and (14), β_{HQIV} depends on $(t_{\text{wall}}, T_{\text{obs}})$ only, not on individual photon frequencies. All photons reaching the same observer get the same imprint — a property formally stated as `predictedBirefringence_frequency_independent`, and exactly the PR4 finding across 30–353 GHz. A per-photon ω^2 mechanism would predict $\sim 140\times$ amplification across that range and is ruled out; HQIV does not predict that. The blackbody Planck spectrum smears in photon energy; the β value does not.

Isotropy. m_{prop} is a scalar. β_{HQIV} has no preferred direction, matching the PR4 isotropy null result — formal statements: `predictedBirefringence_isotropic` and `nearPole_predictedBirefringence_isotropic`.

8. FRIEDMANN SCALING RECOVERED, NOT IMPORTED

In a radiation-dominated FLRW cosmology $H \propto T^2/M_{\text{Pl}}$ and so the Hubble time is $t_H = 1/H \propto M_{\text{Pl}}/T^2$. This is famously the quadratic growth law.

In HQIV, identifying one propagation shell with the bundle of new lattice simplices at the observation depth (the same identification that delivered (14)) gives the propagation-shell duration $\sim t_{\text{Pl}} \mathcal{N}(m_T) \sim t_{\text{Pl}} (T_{\text{Pl}}/T)^2 = M_{\text{Pl}}/T^2$ in Planck units.

Theorem 8.1 (Friedmann scaling recovered). *Under the propagation-shell identification of §7,*

$$t_H^{\text{HQIV}}(T) \sim t_{\text{Pl}} \mathcal{N}(m_T) \sim M_{\text{Pl}}/T^2.$$

The same stars-and-bars count (1) that delivers the finite-mode Kirchhoff law also delivers Friedmann's $H \propto T^2$. No Einstein equation has been used.

9. LEAN FORMALISATION

The full derivation chain is sorry-free against mathlib in the following modules.

Mode-count axiom. (Hqiv/Geometry/OctonionicLightCone.lean): `latticeSimplexCount_eq`, `cumLatticeSimplexCount_closed`, `latticeSimplexCount_pos`.
Finite-mode Kirchhoff law. (Hqiv/Physics/HorizonBlackbodySpectrum.lean): `shellOmega` ((3)), `nBose`, `nBose_pos` ((4)), `shellModeMultiplicity` (equal to $\mathcal{N}(m)$ in the new-modes basis), `shellSpectralEnergy` ((5)), `blackbodyEnergyDensity` ((6)), `blackbodyEnergyDensity_pos_of_le`, `blackbodyEnergyDensity_mono_UV`, `blackbodyEnergyDensity_mono_IR` (Theorem 5.4), `vacuumZeroPointEnergy`, `blackbodyEnergyDensityWithVacuum` (Theorem 5.8), `planckUVCutoff` (Planck pole), `referenceIRCutoff` (lock-in IR cutoff).
Stefan and Wien follow-on. (Hqiv/Physics/HorizonBlackbodyStefan.lean, HorizonBlackbodyWienDisplacement.lean, HorizonBlackbodyGreybody.lean): the T^4 scaling and Wien displacement bound are corollaries of Theorem 5.4 on the finite shell list.
Proton-mass test. (Hqiv/Physics/DerivedNucleonMass.lean, QuarkMetaResonance.lean): `protonAnchorMass` (= 938.272), `protonMassFromMetaHarmonics_MeV`, `derivedProtonMass`, `proton_mass_from_shared_harmonics` ((9)), `derivedProtonMass_pos`.
Birefringence test. (Hqiv/Cosmology/CosmologicalShellLadder.lean): `latticeSimplexCount_as_shell_coarsening` (cast of (1)), `m_prop_HQIV_internal` ((14)), `m_prop_HQIV_internal_eq`, `NearPoleObservation.predictedBirefringence` ((16)), `nearPole_cmb_shell_ladder_pass` (numerics within PR4 errors), `predictedBirefringence_isotropic`, `nearPole.predictedBirefringence_isotropic`, `predictedBirefringence_frequency_independent`, `predictedBirefringence_zero_of_no_shell_crossing`, `nearPole.predictedBirefringence_lab_bound` (any lab-scale signal is undetectable).
Wall-clock age input. (Lean snapshot in Hqiv/Geometry/UniverseAge.lean): `age_wall_clock_Gyr_paper` (= 51.2), `age_apparent_Gyr_paper` (= 13.8), `wallClockAgeHomogeneous`, `HQVM_lapse`. The numerical value 51.2 Gyr is the output of the HQIV-modified CLASS implementation in `class_hqiv_patches/source`, not a constant fed in by hand; the Lean module records the snapshot for proof reuse.

10. DISCUSSION: WHERE NORMALISATION WENT

In the standard derivation of the Planck spectrum, a UV regulariser is implicit (a cavity cutoff, dimensional reg, or simply trust in the Bose factor); an IR regulariser is also implicit (the cavity volume). Renormalisation theory then constructs observables independent of these regularisers. This is conceptually expensive — the cutoffs are tools, not physics.

HQIV trades this for two combinatorial inputs. The discrete null lattice provides a structural UV cutoff at $m = 0$ (Planck pole); the lock-in reference shell provides a structural IR cutoff at $m = M$. Both cutoffs are physical choices, not regulators: moving them changes observables in a calculable way, captured by the monotonicity statements of Theorem 5.4. The total mode budget at any pair of cutoffs is the closed-form hockey-stick sum (2).

What is gained:

- No UV catastrophe — the high-frequency modes do not exist.
- No IR divergence — the low-frequency modes are finite in count.
- No renormalisation procedure — the sums are finite as written.
- Under the `proton_lockin` scale-witness convention, one non-trivial quantitative prediction ($\beta = 0.379^\circ$ at present epoch) survives the data at -0.40σ and discriminates wall-clock from apparent age internally.
- Friedmann scaling recovered as a consequence (Theorem 8.1), not imported.

What is *not* claimed:

- HQIV is not claimed to be independent of every scale anchor. The proton mass at $m = 4$ uses the theory’s internal constituent-plus-binding readout, with the observed value 938.272 MeV serving as the active scale witness. Substituting an alternative witness (`codata.alpha`, `cmb.now`) would re-pin every dimensionful HQIV output; no such cross-witness consistency demonstration is given here.
- The shell index $m = 4$ is not a two-axiom output. It is the QCD-onset identification, and the constituent-plus-binding closure at that shell to 938.272 MeV is a non-trivial HQIV calculation, but the placement of $m = 4$ itself is a modelling choice.
- The wall-clock age 51.2 Gyr is an output of the HQIV-modified CLASS implementation (`class_hqiv` tree, ADM lapse, φ -dependent evolution), not a free parameter, but neither is it strictly forced by the two axioms alone — the `class_hqiv` background carries its own modelling choices on top of the lattice.
- One further piece of the β chain is an explicit identification, not a strict consequence of the two axioms: “one propagation shell = one bundle of new lattice simplices at the observation depth” (the same construction that gives Friedmann’s T^2 scaling).

The proper reading of this paper is therefore: HQIV’s combinatorial mode-count axiom delivers Kirchhoff’s law on a finite shell list without any regularisation; under the standing `proton.lockin` scale-witness convention, combining the HQIV wall-clock age from `class_hqiv` with the lattice mode count $\mathcal{N}(m)$ under the propagation-shell identification predicts $\beta_{\text{HQIV}} = 0.379^\circ$, surviving the PR4 data to -0.40σ . Stronger claims of full independence require either a derivation of $m = 4$ and t_{wall} from the two axioms or a cross-witness consistency demonstration; both are open theoretical targets.

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